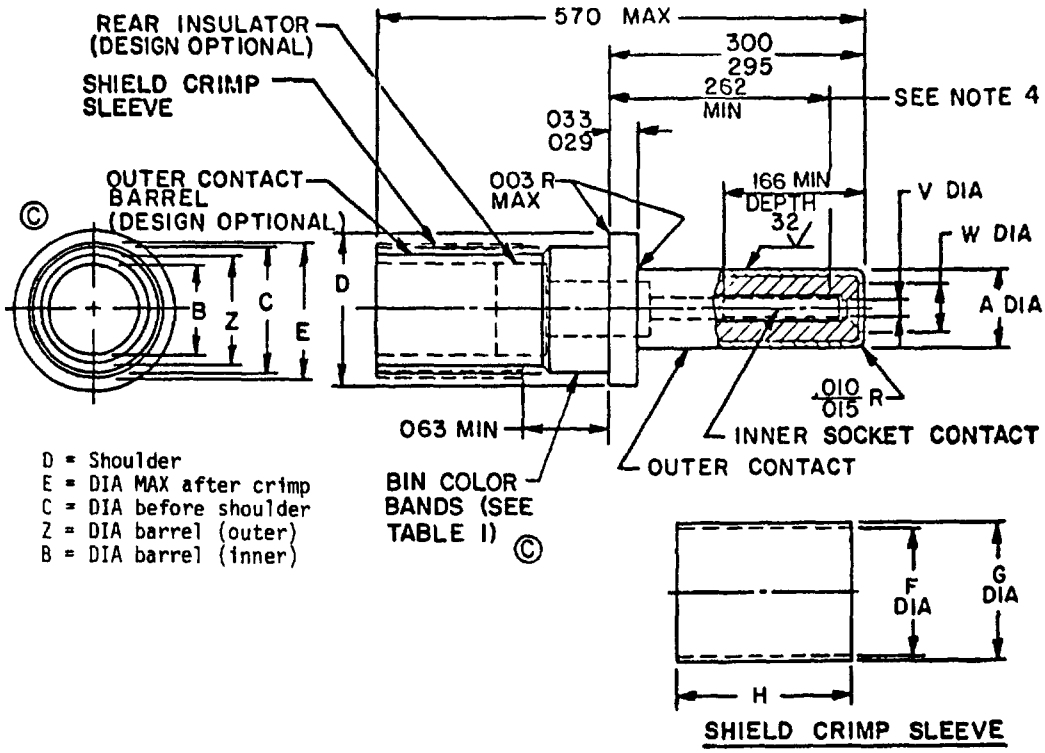
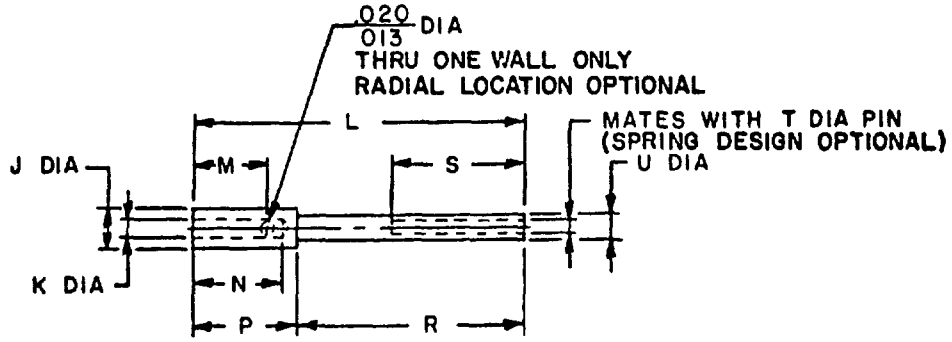
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR,
PIN, CRIMP REMOVABLE, SHIELDED, SIZE 12
(FOR MIL-C-38999 SERIES I, II, III AND IV CONNECTORS)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029





INNER SOCKET CONTACT

© Denotes Changes

FIGURE 1 Pin contacts

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.

BIN code	A Dia.	B Dia Min.	C Dia	D Dia	E Dia. Max.	F Dia Min	G Dia Max	H	J Dia. Max	K Dia Min	L REF
211	095 .093	090	151 .148	.182 .179	156	.127	169	125 115	052	0225	363
409		108				144				.0225	
410		108				.144				.0355	
411		.090				.127				.0355	
412		090				.127	169			0225	
413		117				156	174			0270	
414		090				.144	169			0225	
415	095 .093	.108	151 148	182 179	156	156	174	125 115	052	0355	363

BIN code	M	N Min	P	R	S Min	T	U	V Dia	W Dia	Z Dia Max
211	103 .096	112	146 .140	222 219	156	0205 0195	.035 .033	027 025	058 055	110
409										.127
410										127
411										110
412										.110
413										136
414										110
415	103 096	112	146 140	.222 219	156	.0205 0195	035 033	027 025	058 055	127

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
003	.08	.025	.64	.058	1 47	110	2 79	.144	3 66	.179	4.55
.010	.25	.027	.69	.063	1 60	112	2 84	146	3 71	182	4 62
.013	.33	.029	.74	090	2 29	.115	2 92	148	3 76	.219	5.56
.015	.38	.033	84	.093	2 36	.117	2.97	151	3 84	222	5 64
0195	.495	.035	.89	.095	2.41	125	3 18	156	3.96	.262	6 65
020	.51	.0355	.902	.096	2.44	127	3.23	.166	4.22	295	7 49
.0205	.521	052	1 32	.103	2.62	.136	3.45	169	4 29	.300	7 62
.0225	.572	055	1 40	108	2.74	140	3 56	.174	4 42	.363	9.22
										570	14.48

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 4 mm.
3. Dimensions shown apply after plating
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the inner contact spring. Provision for clearance hole shall be provided.
5. Crimp deformation The maximum diameter over the crimped portion of the shield crimp sleeve shall not exceed E diameter

FIGURE 1. Pin contacts - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration: See figure 1 and table I.

Mating contacts: MIL-C-39029/27 and MIL-C-39029/75.

Tools: See table II.

TABLE I. Design characteristics.

BIN code	Color bands			Cable accommodated	Contact cavity size	Type	Class
	1st	2nd	3rd				
211	Red	Brown	Brown	1/2/ M17/119-RG174 M17/113-RG316 M17/094-RG179 Times AA3248 Teledyne 11299 Thermax 75-738-BCCWXE Tensolite 30888/L707YX-1 Haveg 8100207	12	D	B
409	Yellow	Black	White	1/ M17/095-RG180 Raychem 9527D1514-2L Raychem 9528A1318 Microdot 293-3922			
410	Yellow	Brown	Black	1/ Microdot 250-4070			
411	Yellow	Brown	Brown	1/ Raychem 48-502 & 5022E5111			
412	Yellow	Brown	Red	1/ Raychem 48-950 & 9530D5117			
413	Yellow	Brown	Orange	1/ Raychem 7624D1311 Raychem 9527A1318			
414	Yellow	Brown	Yellow	1/ Gore GWN1159A			
415	Yellow	Brown	Green	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	↓	↓	↓

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

© TABLE II. Tools.

BIN code	Inner contact		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Positioner		
211, 409, 410, 411, 412, 413, 414, 415	M22520/2-01	M22520/2-34	M22520/31-01	M22520/31-02	M81969/8-09 or M81969/14-04	M81969/8-10 or M81969/14-04

© Cable to contact information: See table III

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.

© TABLE III Cable to contact information

BIN code	Cable accommodated	Inner contact tool selector setting no
211	M17/119-RG174	4
	M17/113-RG316	5
	M17/094-RG179	3
	1/ 2/ Times AA3248	4
	Teledyne 11299	4
	Thermax 75-738-BCCWXE	4
	Tensolite 30888/L707YX-1	4
	Haveg 8100207	4
409	M17/095-RG180	3
	1/ Raychem 9527D1514-2L	5
	Raychem 9528A1318	4
	Microdot 293-3922	3
410	1/ Microdot 250-4070	4
411	1/ Raychem 48-502 & 5022E5111	4
412	1/ Raychem 48-950 & 9530D5117	3
413	1/ Raychem 7624D1311	5
	Raychem 9527A1318	5
414	1/ Gore GWN1159A	4
415	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	4

1/ Or equivalent
2/ High tensile strength copper alloy wire

Contact resistance: See table IV.
Test current:
Inner contact - 1 ampere.
Outer contact - 12 amperes.

Low signal level contact resistance (inner contact only)· See table V.

Contact engagement and separation forces (inner socket contact only) The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197 except the diameters shall be as specified in the following, and surface roughness shall not exceed 3 microinches. Provision for clearance hole shall be provided.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.0205 +.0002 -.0000	NA	NA	12	14	NA
.0195 +.0000 -.0002	0.5	0.4	NA	NA	NA

Dielectric withstanding voltage (applied between inner and outer contact)·
Test voltage·
At sea level - 1000 Vac rms.
At 50,000 feet - 250 Vac rms.

Tensile strength (inner and outer contact crimp joint) See table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.

TABLE IV. Contact resistance.

BIN code	Cable accommodated	Maximum voltage drop (millivolts)						Maximum average voltage drop
		25° +3° -0°C		2/ 25° +3° -0°C		200° +3° -0°C		
		Inner contact	Outer contact	Inner contact	Outer contact	Inner contact	Outer contact	
211	M17/119-RG174	55	85	66	102	94 4/	145 4/	NA
	M17/113-RG316	55	75	66	90	94	128	
	M17/094-RG179	120	70	144	84	204	119	
	1/3/ Times AA3248	170	150	204	180	290	255	
	1/3/ Teledyne 11299	↓	↓	↓	↓	↓	↓	
	1/3/ Thermax 75-738-BCCWXE							
	1/3/ Tensolite 30888/L707YX-1							
	1/3/ Haveg 8100207							
409	M17/095-RG180	120	60	144	72	204	102	
	1/ Raychem 9527D1514-2L					204 5/	102 5/	
	1/ Raychem 9528A1318					204 5/	102 5/	
	1/ Microdot 293-3922					204	102	
410	1/ Microdot 250-4070	↓		↓		204	102	
411	1/ Raychem 48-502 & 5022E5111	55		66		94 5/	102 5/	
412	1/ Raychem 48-950 & 9530D5117	120		144		204 5/	102 5/	
413	1/ Raychem 7624D1311	55		66		94 5/	102 5/	
	1/ Raychem 9527A1318	120		144		204 5/	102 5/	
414	1/ Gore GWN1159A	120		144		204	102	
415	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	↓	66	↓	94 4/	102 4/	↓

1/ Or equivalent.
2/ After conditioning.
3/ High tensile strength copper alloy wire.
4/ 85° +3° -0°C.
5/ 115° +3° -0°C.

© TABLE V. Low signal level contact resistance (inner contact only) and tensile strength.

BIN code	Cable accommodated	Maximum contact resistance (milliohms)		Tensile load (pounds) (minimum)	
		Initial	After conditioning	Inner contact	Outer contact
211	M17/119-RG174	55	66	15.0	15.0
	M17/113-RG316	55	66	10 0	15.0
	M17/094-RG179	120	144	3.5	15.0
	1/2/ Times AA3248	170	204	7.0	35.0
	1/2/ Teledyne 11299	↓	↓	↓	↓
	1/2/ Thermax 75-738-BCCWXE				
	1/2/ Tensolite 30888/L707YX-1				
	1/2/ Haveg 8100207				
409	M17/095-RG180	120	144	3.5	20.0
	1/ Raychem 9527D1514-2L			6.0	
	1/ Raychem 9528A1318			6.0	
	1/ Microdot 293-3922			3.5	
410	1/ Microdot 250-4070	↓	↓	4.0	
411	1/ Raychem 48-502 & 5022E5111	55	66	10.0	
412	1/ Raychem 48-950 & 9530D5117	120	144	3.5	
413	1/ Raychem 7624D1311	55	66	6.0	
	1/ Raychem 9527A1318	120	144	6.0	
414	1/ Gore GWN1159A	120	144	4.0	
415	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	66	15.0	↓

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.

© Vibration· Test condition VI, letter J, per method 2005 of MIL-STD-1344 Except directions shall be longitudinal and perpendicular Time duration shall be 8 hours in each direction

© Shock (specified pulse) Test condition D, per method 2004 of MIL-STD-1344

© Shipping· Figure 1 illustrates final assembly, parts are to be shipped unassembled in a suitable container Manufacturers recommended assembly instructions are to be shipped with the unit package.

QPL evaluating activity Defense Electronics Supply Center (DESC-E), Dayton, OH 45444

Military part number See table VI

© TABLE VI Military part number.

BIN code	Military part number
211	M39029/28-12A
409	M39029/28-12B
410	M39029/28-12C
411	M39029/28-12D
412	M39029/28-12E
413	M39029/28-12F
414	M39029/28-12G
415	M39029/28-12H

Custodians.
Army - CR
Navy - AS
Air Force - 85

Review activities·
Army - AR
Navy - EC, MC, OS
Air Force - 11, 15, 17, 99
DLA - ES

Preparing activity·
Air Force - 85

(Project 5935-3051-6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 73236 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

AFALD/PTS
Gentile AF Station Dayton, OH 45444

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:34Z
Check the source to verify that this is the current version before use.